Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	LegalEase: AI-Powered Legal Document Generator
Maximum Marks	1 Mark

This document outlines how the current LegalEase solution can be scaled to handle larger user bases, increased document-generation workloads, and extended features. The plan is based on the current architecture using Streamlit for the frontend, FastAPI for the backend, Gemini 1.5 Pro for AI generation, and document-formatting modules.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	The current solution is documented for local deployment using FastAPI and Streamlit.	Scaling to a larger production user base would require a hosted production deployment architecture.	High
2	Document generation depends on the Gemini API and configured API keys.	Higher request volumes increase dependency on external AI-service availability and capacity.	High
3	The current project focuses on local document generation and export.	Managing large numbers of user-specific generated documents would require scalable storage ar	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Streamlit frontend and FastAPI backend are designed and documented for local deployment.	Deploy the frontend and backend using production/cloud infrastructure; use scalab
2	Data Storage	The documented core workflow focuses on generated documents and downloadable TXT, DOCX and PDF outputs.	Introduce persistent storage and a database for users, document met
3	Performance	Gemini 1.5 Pro is used for generation and the application performs formatting/export after generation.	Use asynchronous processing, request queuing, caching where appro
4	Security	The Gemini API key is stored through environment configuration, and the project emphasizes secure handling of sensitive data.	Add production authentication, authorization, secrets management, encrypted stora

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Production deployment of the FastAPI backend and Streamlit frontend with secure environment configuration.	0–3 months	Improves availability and establishes a production-ready d
Phase 2	Persistent database and scalable document storage for users, document metadata and generated outputs.	3–6 months	Supports larger data volumes and easier document manag
Phase 3	Authentication, role-based access, audit logging and stronger privacy/security controls.	6–9 months	Improves protection and controlled access for sensitive leg
Phase 4	Advanced legal-document analysis, integration with legal databases, multilingual capabilities and personalized recommendations.	9–12 months	Expands the platform's usefulness beyond basic d

Source basis: LegalEase project documentation. The project describes FastAPI + Streamlit architecture, Gemini 1.5 Pro integration, local deployment, TXT/DOCX/PDF export, secure API-key configuration, and future enhancements such as deeper contract analysis, legal-database integration and personalized recommendations.